CMT 212: CLOUD COMPUTING ESSENTIALS

Topic 1: Introduction to Cloud Computing

By

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WHAT IS CLOUD COMPUTING?

Definition (NIST 2011):

- "A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources..."
- Cloud computing is a service delivery model for accessing computing resources over the internet without owning physical infrastructure.

KEY CHARACTERISTICS (NIST)

1. On-Demand Self-Service
2. Broad Network Access
3. Resource Pooling
4. Rapid Elasticity
5. Measured Service

HISTORICAL EVOLUTION OF CLOUD COMPUTING

- **1960s:** John McCarthy proposes utility computing.
- **1970s:** IBM introduces mainframe virtualization.
- **1999:** Salesforce launches SaaS CRM.
- **2006:** Public release of AWS (EC2, S3).
- **2010s:** Rise of containers, serverless, Kubernetes.
- **2020s:** Multi-cloud, edge, sustainable cloud.

HISTORICAL EVOLUTION OF CLOUD COMPUTING

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1.1 Historical Context

1960s	1960s	2000s	2010s - Present
John McCarthy	Emerarsization VMware	**Cloud Emergencce**	**Cloud Maturity**
• John McCarthy envisioned computing as a public utility • IBM's time-sharing model as early cloud-like usage	• Emergod VMwarer enables abstraction of hardware • Virtual Machines (VMs) allowed better resure utilization and flexibility	• Amazon launched AWS (2006) starting with S3 and EC2 • Google and Microsoft followed with GCP and Azure	• Containers (e.g. Docker) and orchestration (Kubernetes) • Serveriess computing, edge computing, and multi-cloud strategies

1.2 Definitions

A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources...

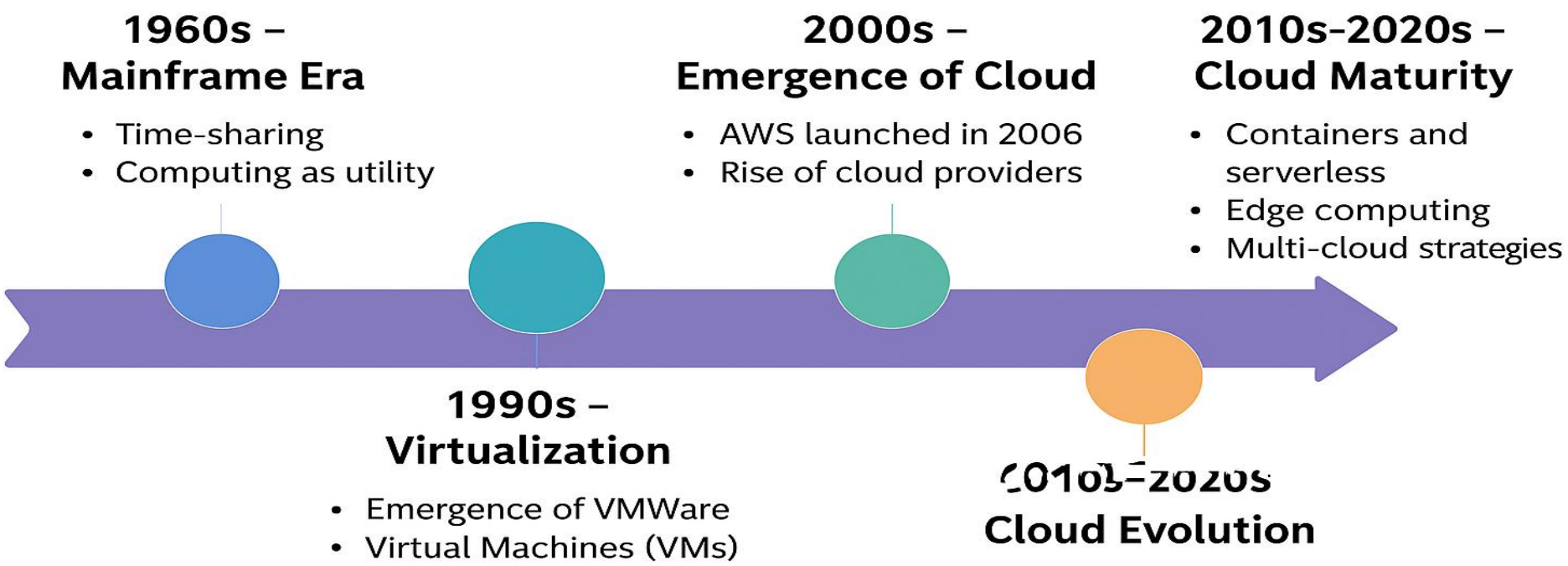
Key Cloud Characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

1.3 Evolution and Current Landscape

- Cloud Native Applications: Microservices, Devop, Ci/CD
- Security as a Priority: Shift to Zero Trust Architecture/
- Emergence of FinOps: Optimizing cloud financial mangement
- Sustainability Goals: Carbon-aware computing and green data centers

HISTORICAL EVOLUTION OF CLOUD COMPUTING



BENEFITS OF CLOUD COMPUTING

1. Cost Efficiency

- **Lower Capital Expenditure**: Cloud computing eliminates the need to invest heavily in physical hardware, servers, and infrastructure.
- **Pay-as-you-go model**: Organizations only pay for what they use computing power, storage, and bandwidth, making it ideal for businesses of any size.
- **Reduced Operational Costs**: No need to maintain or upgrade hardware, manage cooling, or employ large IT staff for infrastructure.
- **Predictable Billing**: Most cloud services provide clear pricing structures, helping with budgeting and cost control.

BENEFITS OF CLOUD COMPUTING

2. Scalability

- **On-Demand Resources**: Businesses can increase or decrease computing resources (like storage or CPU) instantly as demand changes.
- **Supports Business Growth**: Startups can start small and scale up as they grow without needing to overhaul infrastructure.
- **No Downtime for Upgrades**: Scaling can be done dynamically, without affecting system availability or performance.

BENEFITS OF CLOUD COMPUTING

3. Accessibility

- **Access Anytime, Anywhere**: Users can access applications and data via the internet using any device (laptop, smartphone, tablet).
- **Remote Collaboration**: Teams across different locations can work together in real time, enhancing productivity.
- **Platform Independence**: Works across different operating systems and devices, enabling flexibility for users.

BENEFITS OF CLOUD COMPUTING

4. Disaster Recovery

- **Built-in Redundancy**: Cloud providers replicate data across multiple locations, ensuring it’s safe even if one site fails.
- **Quick Recovery**: Cloud-based backups and recovery tools allow for faster restoration of services in case of data loss, hardware failure, or cyberattack.
- **Affordable Protection**: Even small businesses can access enterprise-level disaster recovery solutions without the high costs.

BENEFITS OF CLOUD COMPUTING

5. Faster Innovation

- **Rapid Deployment**: New applications or services can be launched in minutes, not weeks or months.
- **Access to Latest Tech**: Users get immediate access to the latest tools, AI models, machine learning platforms, and more without manual installations.
- **Experimentation and Prototyping**: Developers can test and deploy ideas quickly, fail fast, and iterate, speeding up innovation cycles.

RISKS AND CHALLENGES

- Security (data breaches, misconfigurations)
- Compliance with GDPR, HIPAA, etc.
- Downtime risks
- Vendor Lock-in

CLOUD SERVICE TYPES (OVERVIEW)

- **IaaS** – Infrastructure-as-a-Service
- **PaaS** – Platform-as-a-Service
- **SaaS** – Software-as-a-Service

REAL-WORLD APPLICATIONS

1. Education: Google Classroom, MS Teams

- **Google Classroom** and **MS Teams** rely on cloud infrastructure to provide remote learning platforms.
- Teachers and students can **access assignments, lectures, and grades** anytime, anywhere.
- Cloud enables **real-time collaboration**, video conferencing, file sharing, and secure data storage.
- Ideal for **hybrid and distance learning**, especially during disruptions like COVID-19.

REAL-WORLD APPLICATIONS

2. Healthcare: Cloud-based EHR systems

- **Electronic Health Records (EHR)** systems store patient data securely in the cloud.
- Enables **authorized personnel across hospitals and clinics to have real-time access to medical histories**.
- Enhances **telemedicine**, **data analytics**, and **AI-assisted diagnosis**.
- Offers **data backup**, compliance (e.g., HIPAA), and **disaster recovery** in healthcare settings.

REAL-WORLD APPLICATIONS

3. Finance: FinTech on AWS/GCP

- Cloud platforms like **AWS (Amazon Web Services)** and **GCP (Google Cloud Platform)** power modern **FinTech applications**.
- Used for **mobile banking, digital wallets, fraud detection**, and **algorithmic trading**.
- Benefits include **high scalability**, **security compliance**, and **real-time analytics**.
- Startups use cloud to quickly deploy and scale financial services with low upfront investment.

REAL-WORLD APPLICATIONS

4. Entertainment: Netflix on AWS

- **Netflix** uses AWS for **streaming content** to millions of users globally.
- Stores massive volumes of video content, processes viewer preferences, and adjusts quality based on connection.
- Cloud enables **recommendation engines**, **global content delivery**, and **high availability**.
- Supports **scaling** during spikes (e.g., new releases) and **reduces downtime**.

REAL-WORLD APPLICATIONS

5. Smart Cities: IoT-cloud systems

- Smart cities use cloud-connected **IoT (Internet of Things)** devices for traffic control, energy use, waste management, and surveillance.
- Sensors collect data and send it to the cloud for **analysis, automation, and decision-making**.
- Enables **real-time monitoring**, predictive maintenance, and **efficient city planning**.
- Examples include **smart traffic lights**, **public safety systems**, and **environmental monitoring**.

EMERGING TRENDS IN CLOUD

1. Edge Computing

- **Definition**: Processing data closer to the source (e.g., sensors, devices) instead of sending it to a central cloud.
- **Purpose**: Reduces **latency**, improves **real-time processing**, and decreases **bandwidth use**.
- **Use Cases**: Autonomous vehicles, smart factories, healthcare monitoring devices.
- **Example**: A smart traffic camera processing video data locally for quick decision-making.

EMERGING TRENDS IN CLOUD

2. Serverless Architecture

- **Definition**: Developers write code without managing servers; the cloud provider handles server provisioning and scaling.
- **Benefits**:
- Pay only when the code runs.
- Automatic scaling.
- Faster development and deployment.
- **Use Cases**: Web apps, APIs, background jobs.
- **Example**: AWS Lambda running a function to process uploaded files or trigger notifications.

EMERGING TRENDS IN CLOUD

3. AI/ML-as-a-Service

- **Definition**: Cloud providers offer ready-to-use **Artificial Intelligence (AI)** and **Machine Learning (ML)** tools via APIs.
- **Purpose**: Democratizes AI—anyone can use advanced AI without needing deep expertise or infrastructure.
- **Examples**:
- Google Cloud Vision for image recognition.
- AWS SageMaker for training ML models.
- Microsoft Azure Cognitive Services for language understanding.
- **Use Cases**: Chatbots, recommendation systems, predictive analytics.

EMERGING TRENDS IN CLOUD

4. Green Cloud Initiatives

Definition: Efforts to make cloud computing more environmentally sustainable.

Strategies:

- Using **energy-efficient data centers**.
- Shifting to **renewable energy** sources.
- Optimizing computing workloads to reduce power consumption.

Examples:

- Google Cloud running on **100% renewable energy**.
- Microsoft aiming for **carbon negativity** by 2030.
- **Impact**: Reduces the carbon footprint of digital infrastructure.

SAMPLE QUIZ QUESTIONS

1. List the five NIST characteristics of cloud computing.
2. Who proposed utility computing?
3. Give two examples of SaaS.
4. When was AWS launched publicly?
5. List three benefits and two risks of cloud computing.

SUMMARY

- Cloud computing enables scalable, cost-efficient access to IT resources.
- Rooted in mainframe and virtualization.
- Supports modern apps from AI to IoT.
- Foundation for understanding service and deployment models.

SUMMARY

END OF THE LECTURE